

Django Forms - Complete Coding Guide

Phase 1: HTML Forms

```
<!-- form.html -->
<form>
    Name: <input type="text">
    Age: <input type="number">
    <input type="submit" value="Submit">
</form>
```

Phase 2: GET Requests

```
<!-- form.html -->
<form>
    Name: <input type="text" name="name">
    Age: <input type="number" name="age">
    <input type="submit">
</form>

# views.py
def form(request):
    name = request.GET.get('name')
    age = request.GET.get('age')
    return HttpResponse(f"{name} - {age}")
```

Phase 3: POST Requests

```
<!-- form.html -->
<form method="POST">
    {% csrf_token %}
    <input type="text" name="name">
    <input type="number" name="age">
    <input type="submit">
</form>

# views.py
def form(request):
    if request.method == "POST":
        name = request.POST.get('name')
        age = request.POST.get('age')
        return HttpResponse(f"{name} - {age}")
    return render(request, 'calc/form.html')
```

Phase 4: Django Forms

```
# forms.py
from django import forms

class StudentForm(forms.Form):
    name = forms.CharField()
    age = forms.IntegerField()

# views.py
form = StudentForm()
```

Phase 5: Form Fields

```
class StudentForm(forms.Form):
    name = forms.CharField(max_length=100)
    email = forms.EmailField()
    age = forms.IntegerField()
    website = forms.URLField()
    agree = forms.BooleanField()
```

Phase 6: Widgets

```
class StudentForm(forms.Form):
    name = forms.CharField(
        widget=forms.TextInput(attrs={'class': 'form-control'})
    )

    password = forms.CharField(
        widget=forms.PasswordInput()
    )
```

Phase 7: Validation

```
class StudentForm(forms.Form):
    name = forms.CharField()

    def clean_name(self):
        name = self.cleaned_data['name']
        if len(name) < 3:
            raise forms.ValidationError("Too short")
        return name
```

Phase 8: Error Handling

```
if form.is_valid():
    print("Valid")
else:
    print(form.errors)

<!-- template -->
{{ form.errors }}
```

Phase 9: CSRF Protection

```
<form method="POST">
    {% csrf_token %}
    {{ form }}
    <input type="submit">
</form>
```

Phase 10: Styling Forms

```
{{ form.as_p }}
{{ form.as_table }}
{{ form.as_ul }}
```

Phase 11: ModelForm

```
# models.py
class Student(models.Model):
    name = models.CharField(max_length=100)
    age = models.IntegerField()

# forms.py
class StudentForm(forms.ModelForm):
    class Meta:
        model = Student
        fields = '__all__'
```

Phase 12: Save Data

```
if form.is_valid():
    form.save()
```

Phase 13: Update Data

```
student = Student.objects.get(id=1)

form = StudentForm(
    request.POST or None,
    instance=student
)

if form.is_valid():
    form.save()
```

Phase 14: Delete Data

```
student = Student.objects.get(id=1)
student.delete()
```

Phase 15: File Uploads

```
# models.py
photo = models.ImageField(upload_to='photos/')

# forms.py
class PhotoForm(forms.ModelForm):
    class Meta:
        model = Student
        fields = ['photo']
```

Phase 16: Formsets

```
from django.forms import formset_factory

StudentFormSet = formset_factory(
    StudentForm,
    extra=3
)
```